

Pega CSA

Certified System Architect — Interview Question Bank

182 questions with concise answer notes | standard concepts + scenario-based

This document covers the full range of topics tested for the Pega Certified System Architect role: platform fundamentals, case management, data model, UI, process/flows, declarative rules, integration, security, reporting, and delivery best practices — plus three sections of scenario-based questions that hiring managers commonly use to test applied design thinking. Answer notes are intentionally concise; treat them as talking points to expand on in your own words.

Contents

1. Pega Platform Fundamentals (Q1–Q15)
2. Case Management (Q16–Q30)
3. Data Model, Classes & Properties (Q31–Q45)
4. User Interface (UI) (Q46–Q60)
5. Process, Flows & Automation (Q61–Q75)
6. Decisioning & Declarative Rules (Q76–Q85)
7. Integration (Q86–Q95)
8. Security (Q96–Q105)
9. Reporting (Q106–Q115)
10. Application Development & Best Practices (Q116–Q130)
11. Scenario-Based Questions — Case Management & Process (Q131–Q142)
12. Scenario-Based Questions — Data, UI & Rules (Q143–Q155)
13. Scenario-Based Questions — Integration, Security & Deployment (Q156–Q167)
14. Rapid-Fire Conceptual Questions (Q168–Q182)

1. Pega Platform Fundamentals

Q1. What is Pega and what problem does it solve?

Answer: Pega is a low-code BPM/CRM platform for building enterprise applications by modeling business processes as rules rather than writing code. It unifies case management, decisioning, and automation.

Q2. What does 'model-driven, no-code/low-code' mean in Pega?

Answer: Applications are built by configuring reusable rules (visual models) instead of hand-coding, so business and IT can collaborate and change apps quickly.

Q3. What is a rule in Pega?

Answer: A rule is the fundamental building block/instance stored in the PegaRULES database that defines application behavior — flows, activities, properties, UI, decisions, etc.

Q4. What is the PegaRULES database?

Answer: The relational database that stores all rules, work objects, data, and system data. Rules live in the rules schema; work/data live in the data schema.

Q5. Explain the class hierarchy and its purpose.

Answer: Classes organize rules and define reuse/inheritance. Higher classes are more abstract/reusable; lower classes are specific. It enables rule resolution and reuse.

Q6. Differentiate pattern inheritance vs directed inheritance.

Answer: Pattern inheritance walks up the class name by dropping the last segment (dash-separated). Directed inheritance uses the explicitly-defined parent class. Pattern is checked first, then directed.

Q7. What is a class group / work pool?

Answer: A class group ties concrete case classes that share a single database table and a common ID prefix, so related cases store together.

Q8. What is an Application rule?

Answer: It defines the application layer — built-on applications, rulesets in scope, access, and case types. It's the top of the ruleset stack.

Q9. What is a ruleset and a ruleset version?

Answer: A ruleset is a container/named collection of rules for packaging, security, and migration. Versions (e.g., 01-01-01) control which rules are active via the ruleset stack.

Q10. Explain the ruleset stack and its role in rule resolution.

Answer: The ordered list of rulesets/versions in scope for a requestor. Rule resolution searches the stack top-down to find the applicable rule instance.

Q11. What is rule resolution?

Answer: The algorithm Pega uses to pick the single most appropriate rule at runtime, filtering by ruleset stack, class hierarchy, availability, circumstance, date, and applies-to class.

Q12. What is the significance of the 'Applies To' class of a rule?

Answer: It anchors the rule in the class hierarchy, controlling where the rule is available for resolution and reuse.

Q13. What are the main built-on applications every app inherits?

Answer: PegaRULES / Pega Platform base plus Pega frameworks; app-specific implementation is built on an org/enterprise layer following the Enterprise Class Structure.

Q14. Explain the Enterprise Class Structure (ECS) layers.

Answer: Three layers: Implementation (specific app), Division/Framework (shared reusable), and Organization (enterprise-wide). Promotes reuse and specialization.

Q15. What is App Studio vs Dev Studio?

Answer: App Studio is the business-friendly, guided design environment; Dev Studio (Designer Studio) exposes all rules and advanced configuration for developers.

2. Case Management

Q16. What is a case in Pega?

Answer: A case is a unit of work that tracks a business transaction from creation to resolution, e.g., a loan application or service request.

Q17. What is a case type?

Answer: A reusable template/definition describing the stages, steps, data, and lifecycle for a category of cases.

Q18. Explain the case life cycle (stages and steps).

Answer: Work moves through primary stages left-to-right; each stage contains steps (processes/tasks). Stages model the business milestones of the case.

Q19. What is a stage vs an alternate stage?

Answer: Primary stages are the expected happy path; alternate stages handle exceptions/deviations reached via change-stage actions.

Q20. What is a process/flow within a step?

Answer: A sequence of tasks (assignments, decisions, subprocesses) modeling how work is completed within a step.

Q21. What is a child/subcase and when do you use it?

Answer: A case spun off from a parent to handle a portion of work that can progress independently (e.g., separate approvals). Parent can wait on children.

Q22. How do parent and child cases communicate/coordinate?

Answer: Via case dependencies — parent can wait for all/any children to resolve, and data can propagate down at instantiation.

Q23. What is case locking and why does it matter?

Answer: Pega locks a case when opened for update to prevent concurrent edits. Default is pessimistic locking on the case ID.

Q24. Difference between default (pessimistic) and optimistic locking.

Answer: Pessimistic locks on open and holds until released; optimistic locks only at commit and detects conflicts, allowing more concurrency.

Q25. What is a case status (pyStatusWork) and its typical values?

Answer: The work status field: New, Open, Pending, Resolved-Completed, Resolved-Rejected, etc. Used for reporting and routing logic.

Q26. How do you resolve a case?

Answer: Set the status to a Resolved-* value (typically via a Utility/step or the flow reaching an end shape) so the case closes.

Q27. What are goals and deadlines (SLAs) on a case/assignment?

Answer: Service Level Agreements define goal and deadline times with escalation actions (notify, escalate, reassign) when times are exceeded.

Q28. What is a Service Level (SLA) rule?

Answer: A rule defining urgency, goal, deadline, passed-deadline intervals and the escalation activities triggered at each interval.

Q29. What is case-wide vs assignment-level SLA?

Answer: Assignment SLAs govern a single assignment; case-level SLAs govern the whole case's resolution time.

Q30. How does urgency affect work in a worklist/workbasket?

Answer: Higher urgency sorts work higher; SLAs and manual settings adjust urgency to influence prioritization.

3. Data Model, Classes & Properties

Q31. What is a property in Pega?

Answer: A named field on a class that holds a piece of data with a defined type (Text, Integer, Date, etc.) and behavior.

Q32. List the main property modes.

Answer: Single value, Value List, Value Group, Page, Page List, Page Group, and Java Object modes.

Q33. Difference between a Value List and a Value Group.

Answer: Value List is indexed by integer (ordered); Value Group is keyed by a subscript/string (unordered map-like).

Q34. Difference between a Page List and a Page Group.

Answer: Both hold embedded pages; Page List is numerically indexed/ordered, Page Group is keyed by subscript.

Q35. What is a data type in Pega (App Studio term)?

Answer: A data type represents a reusable data object (a class) with its fields and data sources — the App Studio abstraction over a data class.

Q36. What is the difference between a data class and a work/case class?

Answer: Case classes hold work objects that flow through a lifecycle; data classes hold reference/reusable data (e.g., Customer, Address) that supports cases.

Q37. What is a data page (formerly declare page)?

Answer: A rule that loads and caches data on demand into a clipboard page from a source (DB, report, activity, connector) with a defined scope.

Q38. Explain data page scopes: Thread, Requestor, Node.

Answer: Thread = per open case/thread; Requestor = per user session; Node = shared across all users on the server node.

Q39. Editable vs read-only data pages.

Answer: Read-only pages are system-managed and refresh by strategy; editable data pages can be modified and saved back via a savable data page.

Q40. What is a savable data page?

Answer: A data page configured with a save plan so its contents can be persisted back to a system of record.

Q41. What is the clipboard?

Answer: The in-memory workspace of pages and properties for the current requestor/thread; the Clipboard tool inspects it during development.

Q42. What key clipboard pages should a CSA know?

Answer: pyWorkPage (current case), primary page, pxRequestor, OperatorID page, and data pages (D_ prefixed).

Q43. What is a keyed data page and why use it?

Answer: A data page parameterized by keys so a single definition can cache multiple keyed instances (e.g., D_Customer[CustID]).

Q44. What is the difference between .Property and Page.Property references?

Answer: A dot reference is relative to the step/primary page; a fully-qualified reference names the page explicitly.

Q45. What is a calculated (declarative) property vs a stored property?

Answer: Declarative properties are recomputed automatically by declare rules when inputs change; stored properties hold explicitly set values.

4. User Interface (UI)

Q46. What is a section rule?

Answer: A reusable UI building block containing layouts, fields, and controls that can be embedded in harnesses and other sections.

Q47. What is a harness?

Answer: The top-level UI container for a form/portal view (e.g., New, Review, Perform) that arranges sections and layouts.

Q48. What are the common harness types you configure for a case?

Answer: New (create), Perform (work an assignment), Review (read-only), Confirm, and Reopen harnesses.

Q49. What is a control rule?

Answer: Defines how a single field renders and behaves (text input, dropdown, button, etc.) including formatting and actions.

Q50. What is a dynamic layout vs a table layout?

Answer: Dynamic layouts arrange fields responsively (stacked/inline); table/repeating layouts render row-based data lists.

Q51. What are UI skins and their purpose?

Answer: A skin rule centralizes styling (fonts, colors, spacing, responsive breakpoints) so branding is consistent and easy to change.

Q52. What is a portal and what portals exist by default?

Answer: A portal is the user's workspace UI. Defaults include the Case Manager portal, Case Worker portal, and Dev Studio.

Q53. How do you show/hide a field or section conditionally?

Answer: Use a Visible-when condition backed by a when rule or expression on the field/section's visibility setting.

Q54. What is a when rule?

Answer: A reusable boolean rule evaluating a condition (true/false) used for visibility, flow decisions, validation, etc.

Q55. What is client-side vs server-side validation?

Answer: Client-side runs in the browser (required, format) for instant feedback; server-side (validate/edit rules) enforces business rules reliably.

Q56. What is an action set on a UI control?

Answer: A configured event (click, change, blur) mapped to one or more actions (refresh section, run activity, set value, open modal).

Q57. How do you implement a repeating grid bound to a Page List?

Answer: Use a repeating (table) layout sourced from a page list or a data page, mapping each column to a property.

Q58. What is a paragraph rule / correspondence rule?

Answer: Paragraph rules hold reusable rich text; correspondence rules assemble outbound communications (email/letter) using them.

Q59. What is responsive UI in Pega?

Answer: Layouts adapt to screen size using skin breakpoints so the same UI works on desktop, tablet, and mobile.

Q60. What is the difference between Auto-generated and non-auto-generated (custom HTML) sections?

Answer: Auto-generated sections use Pega's managed markup/skinning (recommended); custom HTML is hand-written and harder to maintain/upgrade.

5. Process, Flows & Automation

Q61. What is a flow rule?

Answer: A visual process model of shapes (start, assignments, decisions, subprocess, utility, end) describing how a case advances.

Q62. Name common flow shapes and their purpose.

Answer: Assignment (human task), Decision (branch), Subprocess (call another flow), Utility (automated step), Connector (path), End (terminate).

Q63. What is an assignment shape?

Answer: A point where the flow pauses and routes work to a person or workbasket until acted upon via a flow action.

Q64. What is a flow action?

Answer: A rule defining a user's choice at an assignment — the form and processing that occurs when the user submits it.

Q65. Local flow action vs connector flow action.

Answer: Connector flow actions move the case along a connector to the next shape; local flow actions perform work without leaving the assignment.

Q66. What is a router / how is work assigned?

Answer: Routers (ToWorklist, ToWorkbasket, ToDefaultWorkbasket) determine whether work goes to an operator's worklist or a shared workbasket/queue.

Q67. Difference between a worklist and a workbasket (work queue).

Answer: A worklist is an individual operator's personal to-do list; a workbasket/work queue is a shared pool a team pulls from.

Q68. What is a decision shape backed by (decision rules)?

Answer: It can use a when rule, decision table, decision tree, or map value to pick the outgoing connector.

Q69. Decision table vs decision tree — when to use each.

Answer: Decision tables suit many conditions across columns (matrix); decision trees suit nested/sequential if-then logic branches.

Q70. What is a map value rule?

Answer: A rule that returns a result by looking up one or two input dimensions in a matrix (e.g., zip + amount to a rate).

Q71. What is a subprocess (call vs spinoff)?

Answer: Call waits for the subprocess to finish before continuing; spin-off launches it in parallel and continues immediately.

Q72. What is a ticket in a flow?

Answer: A named marker that lets you jump the flow to a specific point when the ticket is raised (used for exceptions/restarts).

Q73. What is a screen flow?

Answer: A flow that presents a series of assignment screens to one user in sequence (wizard-style) with back navigation.

Q74. What is a straight-through process (STP)?

Answer: A process that completes with no human assignments — fully automated end to end.

Q75. What is an activity and when should a CSA use one?

Answer: An activity is a sequence of low-code steps/methods. Prefer declarative/no-code options first; use activities only when configuration can't achieve the goal.

6. Decisioning & Declarative Rules

Q76. What are declarative rules?

Answer: Rules that automatically enforce relationships/derivations without being explicitly called — Pega recalculates when dependencies change (forward/backward chaining).

Q77. Explain forward vs backward chaining.

Answer: Forward chaining recalculates a target when a source changes; backward chaining computes a value on demand when it is read/needed.

Q78. What is a Declare Expression?

Answer: A declarative rule that automatically computes a target property from a formula whenever its inputs change.

Q79. What is a Declare OnChange rule?

Answer: Runs an activity automatically when a watched property changes value.

Q80. What is a Declare Trigger?

Answer: Runs an activity when instances of a class are saved/updated/deleted in the database (event-driven).

Q81. What is a Declare Constraint?

Answer: Enforces validation rules declaratively, raising messages when conditions are violated.

Q82. What is a Declare Index?

Answer: Exposes embedded/list data into a separate index table so it can be reported on and searched.

Q83. Constraint vs Validate vs Edit Validate — differences.

Answer: Declare Constraint = declarative/continuous; Validate rule = server-side conditional validation invoked in flow/action; Edit Validate = single-property Java-based field check.

Q84. What is a Collection rule?

Answer: A rule that runs an ordered set of steps/rules as a group, useful for orchestrating sequential business logic declaratively.

Q85. What is Pega's decision management (high level for CSA)?

Answer: Configuring business logic that drives next-best-action/decisions via decision rules; deep decisioning (strategies) is more CSSA-level.

7. Integration

Q86. What is a connector vs a service in Pega?

Answer: Connectors let Pega call external systems (outbound); Services let external systems call into Pega (inbound).

Q87. Name common integration types Pega supports.

Answer: REST, SOAP, HTTP, Email, File, JMS/MQ, Database (SQL), and Robotic automation.

Q88. What is a Connect-REST rule?

Answer: Defines an outbound REST integration — endpoint, method, headers, and request/response mapping to Pega properties.

Q89. What is a data source vs a system of record?

Answer: A data source supplies data (DB, connector, report); the system of record is the authoritative external store the source draws from.

Q90. How does a data page typically integrate external data?

Answer: A read-only data page sources from a connector/report and caches it so the app references clean, decoupled data.

Q91. What is response data mapping?

Answer: Mapping fields from the external response payload (JSON/XML) into Pega clipboard properties/data pages.

Q92. What is an integration/data layer and why decouple it?

Answer: Isolating integrations behind data pages/data types so UI and process rules don't depend on integration specifics — easier change and reuse.

Q93. What is a Connect-SOAP rule and when is it used?

Answer: Outbound SOAP web service call using WSDL-defined operations; used for legacy/enterprise SOAP endpoints.

Q94. What is a service package?

Answer: Groups service rules and defines access/authentication/deployment settings for inbound services.

Q95. What role does a virtual mapping / authentication profile play?

Answer: Authentication profiles store credentials/tokens (e.g., OAuth 2.0) reused by connectors for secure outbound calls.

8. Security

Q96. What is an access group?

Answer: Associates an operator with an application, available portals, roles, and ruleset access — the security context for a user.

Q97. What is an operator ID?

Answer: The user account record holding credentials, access group, org unit, work queues, and preferences.

Q98. What is a role and an Access of Role to Object (ARO)?

Answer: A role groups privileges; an ARO maps a role to a class and grants CRUD-level access controls (levels 0–5).

Q99. What is an Access Deny rule?

Answer: Explicitly denies access, overriding grants — used to restrict specific classes/actions for a role.

Q100. What is a privilege?

Answer: A named permission checked by rules (e.g., a flow action) and granted to roles to gate functionality.

Q101. Attribute-based (ABAC) vs Role-based (RBAC) access control.

Answer: RBAC grants by role; ABAC grants dynamically based on properties/attributes of user and object (access control policies).

Q102. What is an authentication service?

Answer: Configuration for how users authenticate — e.g., SSO via SAML 2.0 or OpenID Connect, or basic Pega credentials.

Q103. How do you secure a property/field?

Answer: Use access control policies, encryption for sensitive data, and role/ARO restrictions plus UI visibility conditions.

Q104. What is the difference between authentication and authorization?

Answer: Authentication verifies who the user is; authorization determines what that authenticated user is allowed to do.

Q105. What is client-based / attribute-based access control policy used for?

Answer: To enforce row/field-level security dynamically (read/update/delete/discretionary) based on policy conditions.

9. Reporting

Q106. What is a report definition?

Answer: The primary reporting rule that queries data with columns, filters, sorting, and grouping to produce list or summary reports.

Q107. List vs Summary report — difference.

Answer: List reports show detailed rows; summary reports aggregate (count/sum/avg) and group data with subtotals.

Q108. What are report filters and filter conditions?

Answer: Criteria that limit rows returned; conditions can be combined with AND/OR logic and parameterized.

Q109. How do you report on data stored in embedded pages/lists?

Answer: Expose it with a Declare Index so it lands in a queryable index table the report can reference.

Q110. What are exposed/optimized properties and why optimize?

Answer: Properties stored in dedicated DB columns (vs the BLOB) so they can be filtered/sorted efficiently in reports and searches.

Q111. What is the difference between the BLOB (Storage Stream) and exposed columns?

Answer: The BLOB stores the full serialized case; exposed columns duplicate select properties into real columns for querying/performance.

Q112. How can users access/build reports in Pega?

Answer: Through the Report Browser in the portal, where they create and run report definitions without developer help.

Q113. What is a sub-report / joining data in a report?

Answer: Report definitions can join associated classes or include sub-reports to combine related data sources.

Q114. How do you schedule or distribute a report?

Answer: Configure scheduling to email report output to recipients on a recurring basis via the report's scheduling options.

Q115. What is a chart in a summary report?

Answer: A visual (bar/line/pie) rendering of aggregated report data configured on the summary report's chart tab.

10. Application Development & Best Practices

Q116. What is the Application Development lifecycle in Pega (DCO)?

Answer: Directly Capture Objectives — collaboratively capture requirements as executable specifications, then iteratively build in App Studio/Dev Studio.

Q117. What are case types, personas, and channels in App Studio?

Answer: Case types define work; personas define user roles/access; channels define how users interact (web, mobile, email, chat).

Q118. What is the Guardrails / compliance score?

Answer: Pega's best-practice guardrails; the compliance score measures how well the app follows them (high score = maintainable/upgrade-safe).

Q119. Why avoid custom Java/HTML where possible?

Answer: It lowers the guardrail score, complicates upgrades, and reduces reuse — prefer configured, auto-generated rules.

Q120. What is circumstancing a rule?

Answer: Creating a variant of a rule that applies only under a condition (property value, date, or template) while a base version handles the default.

Q121. What is specialization vs override of a rule?

Answer: Specialization tailors a rule for a lower class/ruleset/circumstance; override replaces the resolved rule higher in the stack.

Q122. What are check-in / check-out and rulesets locking?

Answer: Check-out gives a private editable copy; check-in commits it. Locked ruleset versions prevent edits, enforcing controlled change.

Q123. How do you migrate an application between environments?

Answer: Package rules/data into a product (RAP) rule and import it, or use Pega Deployment Manager for CI/CD pipelines.

Q124. What is a branch ruleset?

Answer: A temporary ruleset for parallel team development that is later merged back into the base ruleset.

Q125. What is the difference between a development, and production ruleset?

Answer: Development rulesets are unlocked/editable in DEV; production rulesets are locked and promoted, preventing runtime changes.

Q126. What tools help you debug in Pega?

Answer: Clipboard, Tracer, Live UI, Rule Inspector, and the log/PLA (Predictive Diagnostic) tools.

Q127. What is the Tracer used for?

Answer: Step-by-step runtime tracing of rule execution (activities, flows, data pages) to diagnose behavior and performance.

Q128. What is Live UI?

Answer: An interactive inspector that maps rendered UI back to the underlying section/harness/rule for quick navigation and edits.

Q129. What is the difference between App Studio case designer and Dev Studio flow?

Answer: Case designer is the business-friendly stage/step view; the Dev Studio flow is the detailed process diagram of the same lifecycle.

Q130. What is a keystore / how are secrets managed (CSA awareness)?

Answer: Sensitive credentials/keys are stored in keystore rules and referenced by authentication profiles rather than hard-coded.

11. Scenario-Based Questions — Case Management & Process

Q131. A loan case must go to a senior manager only when the amount exceeds \$50,000. How would you design this?

Answer: Add a decision shape (or when rule 'AmountGreaterThan50k') after submission that routes high-value cases to a manager approval step/workbasket; low-value ones auto-approve or skip. Use a router to the manager's work queue.

Q132. Customers complain approvals take too long. How do you enforce and escalate turnaround times?

Answer: Attach an SLA to the approval assignment with goal/deadline intervals; on deadline, escalate (raise urgency, notify manager, reassign to a backup workbasket).

Q133. A single order needs three independent departmental approvals happening at the same time. How do you model it?

Answer: Spin off three child cases (or a parallel/split-join process) — one per department — and have the parent wait until all children resolve before proceeding.

Q134. Midway through a case, the customer cancels. How do you stop the flow gracefully from anywhere?

Answer: Use a ticket placed at a cancellation stage/point; when the cancel action fires, raise the ticket to jump the flow to a Resolved-Cancelled end regardless of current position.

Q135. An assignment sits in a shared queue but nobody picks it up. How do you ensure it's handled?

Answer: Route to a workbasket with an SLA that, on deadline, notifies a supervisor and reassigns/raises urgency; optionally use skill-based routing to a qualified team.

Q136. You need a wizard-like multi-screen intake for one user with Back/Next. What do you use?

Answer: A screen flow presenting sequential assignment screens to the same user, enabling back navigation and progress indication.

Q137. A case should auto-complete with no human steps when all data is valid. How?

Answer: Design a straight-through process: validate declaratively, use decision/utility shapes only, and route to Resolved without any assignment shapes.

Q138. Two agents keep overwriting each other on the same case. How do you prevent it?

Answer: Rely on Pega's default pessimistic case locking; if high concurrency is needed on sub-parts, split into child cases or use optimistic locking with conflict handling.

Q139. Business wants an exception path for fraud review outside the normal happy path. How model it?

Answer: Add an alternate stage 'Fraud Review'; use a change-stage action or ticket to move suspicious cases there, then return or resolve based on outcome.

Q140. A manager needs to reassign work from a vacationing employee's worklist. How?

Answer: Use the reassign/bulk transfer feature in the Case Manager portal to move assignments from that operator's worklist to another operator or workbasket.

Q141. Requirement: notify the customer by email at three points in the lifecycle. Low-code approach?

Answer: Add 'Send email' automation steps (utility/step) at those stages using correspondence rules, sourced from case data — no activity needed.

Q142. A process step must run automatically to calculate a fee before the next human task. How?

Answer: Use a Utility (automation) step calling a declarative expression or activity to compute the fee, placed before the human assignment.

12. Scenario-Based Questions — Data, UI & Rules

Q143. A dropdown of countries is the same across many forms and rarely changes. Best design?

Answer: Load it once via a node-level read-only data page sourced from a data type/DB, and reference it from every dropdown for reuse and performance.

Q144. Customer data must be fetched fresh from an external CRM each time a case opens. How?

Answer: Use a thread-scoped read-only data page sourced from a Connect-REST to the CRM, keyed by customer ID, referenced by the case UI.

Q145. A field should appear only when 'Employment Status' = Employed. How?

Answer: Set a Visible-when condition on the field/section using a when rule that checks EmploymentStatus == 'Employed'.

Q146. Total premium must always equal base + tax without anyone clicking calculate. How?

Answer: Use a Declare Expression that sets $\text{Total} = \text{Base} + \text{Tax}$ with forward chaining, so it recalculates automatically whenever Base or Tax changes.

Q147. You must validate that end date is after start date and show a message. Best rule?

Answer: Use a Validate rule (or Declare Constraint) comparing the two dates and adding a property message when the condition fails.

Q148. A grid must show a variable list of line items the user can add/remove. How?

Answer: Bind a repeating (table) layout to a Page List property with add/delete row actions, mapping columns to each item's properties.

Q149. Reports need to filter by an item inside an embedded page list, but can't. Why and fix?

Answer: Embedded/list data lives in the BLOB and isn't queryable; create a Declare Index to expose it to an index table, then report on that.

Q150. Sorting/filtering a report on a property is slow. What do you do?

Answer: Expose/optimize that property into a dedicated database column so the query uses a real column instead of unpacking the BLOB.

Q151. Business wants to rebrand the app's colors and fonts quickly. How?

Answer: Update the skin rule's styling/mixins centrally — no need to touch individual sections since auto-generated UI inherits the skin.

Q152. Only US operators should see a section that EU operators must not. How?

Answer: Control it with a when rule on region plus, for true security, an access control policy/ARO — visibility alone isn't a security guarantee.

Q153. A currency field must show negatives in red and format with commas. How?

Answer: Configure the control's format/type settings and a conditional formatting/style (via when rule or control formatting) for negative values.

Q154. The same customer address section is reused on five forms. Best practice?

Answer: Build one section rule for the address and embed it in all five harnesses/sections — single source, consistent behavior, easy maintenance.

Q155. A property value must trigger a recalculation of downstream discounts only when it changes. Which declarative rule?

Answer: A Declare OnChange (to run logic on change) or Declare Expression (to derive the discount) — Expression if it's a pure calculation.

13. Scenario-Based Questions — Integration, Security & Deployment

Q156. Pega must retrieve inventory levels from a REST API in real time. How?

Answer: Create a Connect-REST rule to the endpoint, invoke it from a read-only data page (keyed by item), and map the JSON response into Pega properties.

Q157. An external system needs to create cases in Pega. What do you configure?

Answer: An inbound service (e.g., Service REST) in a service package with authentication, mapping the incoming payload to a case creation activity/data transform.

Q158. Credentials for an outbound API keep expiring; how manage them cleanly?

Answer: Store them in an authentication profile (e.g., OAuth 2.0) referenced by the connector, and keep secrets in a keystore rather than in the rule.

Q159. A junior developer added lots of custom Java and the app is hard to upgrade. How address it?

Answer: Refactor toward configured/auto-generated rules to raise the guardrail compliance score, improving maintainability and upgrade safety.

Q160. Only supervisors may perform the 'Override Price' flow action. How restrict it?

Answer: Create a privilege, gate the flow action with it, and grant that privilege only to the supervisor role via the role's ARO/privilege settings.

Q161. Agents must only see cases from their own region's customers. How enforce at row level?

Answer: Use attribute-based access control (access control policy) comparing operator region to the case's region for read access.

Q162. You need to move a completed feature from DEV to TEST reliably. How?

Answer: Package the rules/data in a product (RAP) rule and import to TEST, or automate promotion through Pega Deployment Manager pipelines.

Q163. Two teams work on the same application simultaneously without stepping on each other. How?

Answer: Use branch rulesets so each team develops in isolation, then merge branches into the base ruleset with conflict resolution.

Q164. Users authenticate through the company's SSO. What do you set up?

Answer: Configure an authentication service using SAML 2.0 or OpenID Connect so Pega delegates login to the corporate identity provider.

Q165. A connector call intermittently fails and stalls the case. How design for resilience?

Answer: Decouple via a data page with error handling, add retry/timeout settings, and route to an exception/alternate path (or queue for retry) on failure.

Q166. Sensitive SSNs must be masked on screen and protected at rest. What's the approach?

Answer: Mask via control display settings/visibility for most roles, apply access control policies, and enable property-level encryption for storage.

Q167. Production users report a specific rule behaving unexpectedly. How do you diagnose safely?

Answer: Reproduce in a lower environment, use Tracer and the Clipboard to inspect rule resolution and data, and check Live UI for the exact rule rendering.

14. Rapid-Fire Conceptual Questions

Q168. What does 'circumstancing' let you avoid?

Answer: Duplicating whole rules; you create a conditional variant instead of a separate rule for each situation.

Q169. What is the default work ID prefix and how is it set?

Answer: It's defined on the class group (work pool) and appended with a generated number, e.g., W-1001.

Q170. What is pyWorkPage?

Answer: The clipboard page holding the current case/work object being processed.

Q171. Name three ways to set a property value declaratively vs procedurally.

Answer: Declarative: Declare Expression, data transform default, when-driven default. Procedural: activity Property-Set.

Q172. What is a data transform?

Answer: A rule that maps/sets/transforms data between pages/properties without coding — the preferred alternative to activity Property-Set steps.

Q173. Data transform vs activity — which to prefer for data mapping?

Answer: Prefer a data transform; it's declarative, higher on guardrails, and easier to maintain than an activity.

Q174. What is the 'Pega Express' delivery approach?

Answer: An outcome-driven, MLP-first (Minimum Lovable Product) methodology emphasizing quick, iterative business-valued releases.

Q175. What is an MLP (Minimum Lovable Product)?

Answer: The smallest release that delivers real, usable business value — the initial milestone in Pega Express delivery.

Q176. What is the difference between a stage and a step visually in the case designer?

Answer: Stages are the top horizontal phases; steps are the tasks listed vertically within each stage.

Q177. What is a case attachment and how is it added?

Answer: Files/notes/URLs linked to a case; added via attach-content steps, UI attach controls, or automations.

Q178. What is the purpose of a work party?

Answer: Represents people/organizations involved in a case (customer, owner, originator) with roles and contact info.

Q179. What is a specialization by ruleset vs by class?

Answer: Ruleset specialization overrides a rule in a higher ruleset in the stack; class specialization defines it on a more specific class.

Q180. What is the difference between save and commit?

Answer: Save writes to the deferred list within the transaction; commit persists all deferred operations to the database.

Q181. What is Obj-Save vs Obj-Save-Cancel-Save (conceptually)?

Answer: Obj-Save queues an instance to save (deferred); a subsequent commit finalizes it. Understanding deferred vs immediate save is key for data integrity.

Q182. What is the role of the 'Application Overview / Case type landing page'?

Answer: It's where a CSA reviews case types, data model, and personas in App Studio to understand and shape the application.